POLITECNICO DI MILANO

Computing Infrastructures

Course Introduction

Prof. Manuel Roveri
Prof. Gianluca Palermo, Prof. Danilo Ardagna

Prof. Manuel Roveri

- **Full Professor**

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  Email: **manuel.roveri@polimi.it**
  Web: http://roveri.faculty.polimi.it
- Research interests: TinyML, Adaptive AI, Machine learning as a Service, privacy-preserving machine and deep learning
- **Deputy Director of the Department** with Responsibility for Corporate Relations
- **Director of the AI-Tech Research Lab @ Polimi**
- **Co-Founder of DHIRIA**, a Spin-Off of Politecnico di Milano
- Project (co+) coordinator of 25+ projects (1M+ funding)

AI

AI-Tech Research Lab

> Innovation in AI Technologies _

2 Full Professors, 2 Senior Researchers, 8 PhD Students and 2 Junior Researchers

The research activity

Cyber-physical Systems

Artificial Intelligence (Machine learning and deep learning)

The research activity

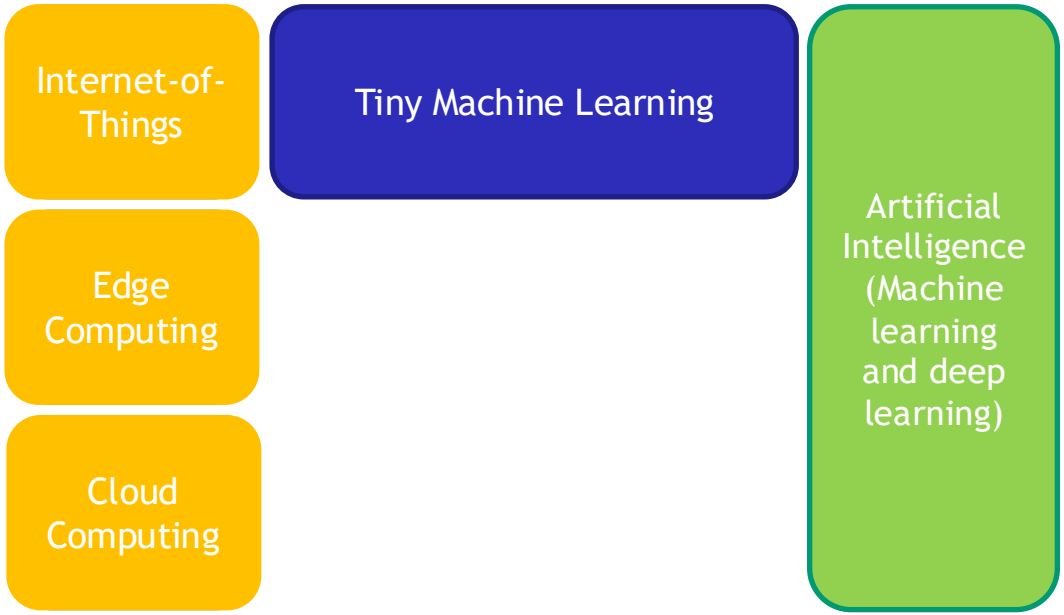
Internet-of-Things

Edge Computing

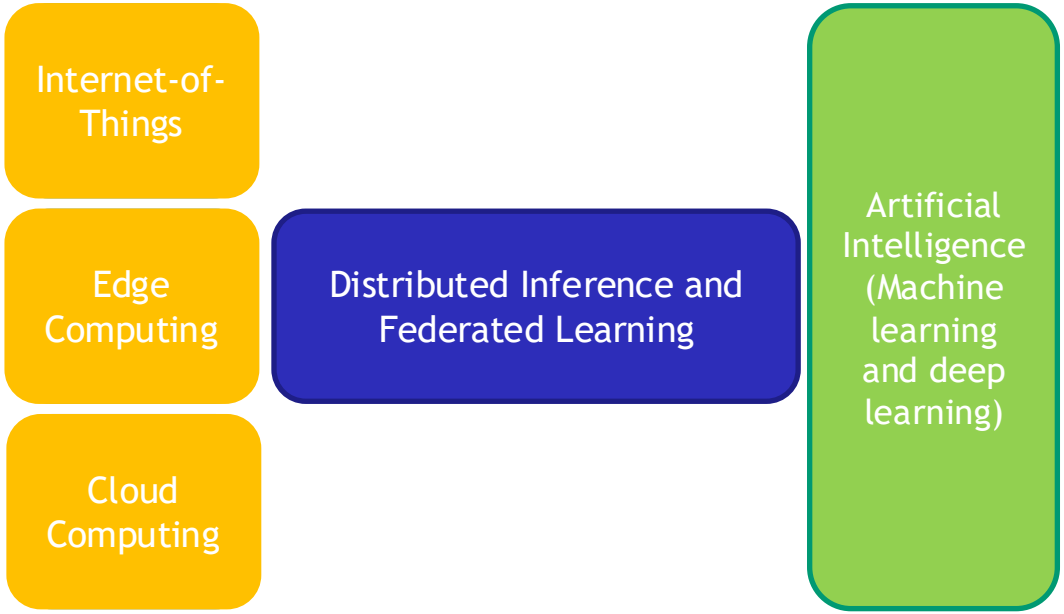
Cloud Computing

Artificial Intelligence (Machine learning and deep learning)

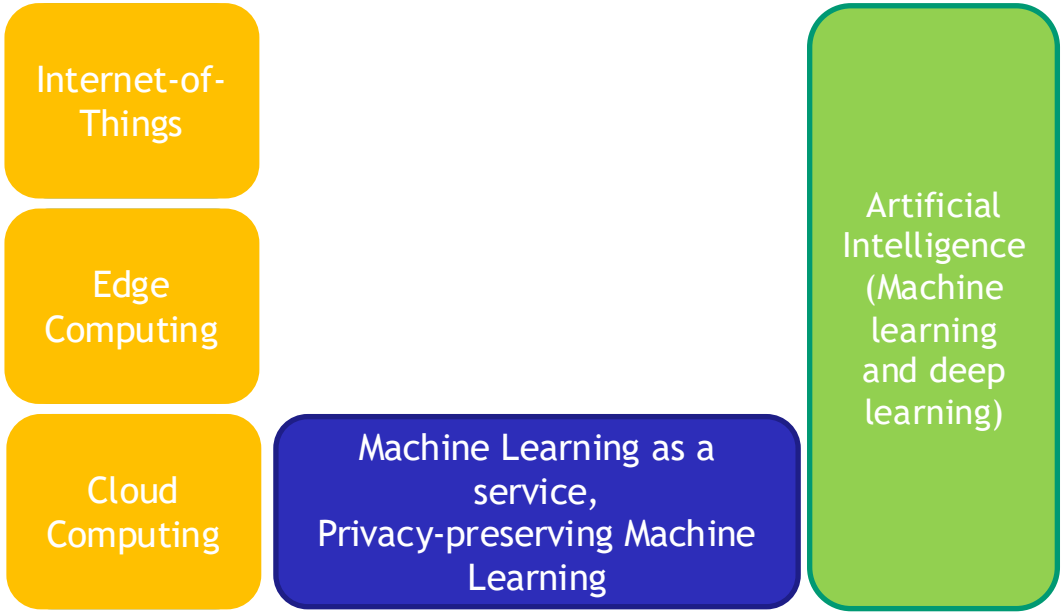
The research activity



The research activity



The research activity



The research activity

Internet-of-Things

Edge Computing

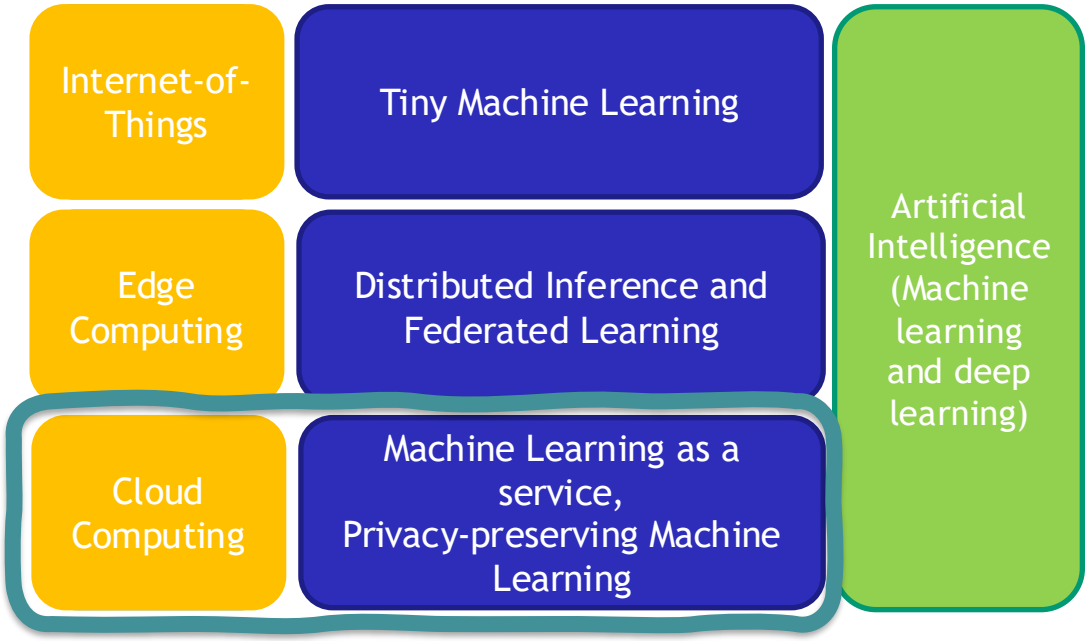
Cloud Computing

Learning in presence of Concept Drift

Intelligent Pervasive Systems

Artificial Intelligence (Machine learning and deep learning)

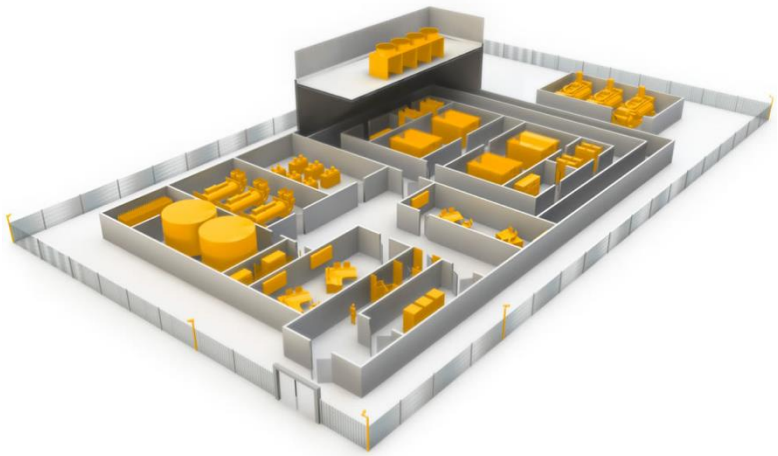
The research activity



Topics of this course!!!

Introduction to the course

- Modern large-scale datacenters require the seamless integration of different components - applications, computation nodes, storage devices, and networks - into one computing infrastructure.



- The course covers the basics of current datacenters architectures, ranging from the analysis of the single components to the global infrastructure.

The topics of the course

A. HW Infrastructures:

- **System-level**: Computing Infrastructures and Data Center Architectures, Rack/Structure;
- **Node-level**: Server (computation, HW accelerators), Storage (Type, technology), Networking (architecture and technology);
- **Building-level**: Cooling systems, power supply, failure recovery

B. SW Infrastructures:

- **Virtualization**: Process/System VM, Virtualization Mechanisms (Hypervisor, Para/Full virtualization)
- **Computing Architectures**: Cloud Computing (types, characteristics), Edge/Fog Computing, X-as-a service
- **Machine and deep learning-as-a-service**

C. Methods:

- **Reliability and availability of datacenters** (definition, fundamental laws, RBDs)
- **Disk performance** (Type, Performance, RAID)
- **Scalability and performance of datacenters** (definitions, fundamental laws, queuing network theory)

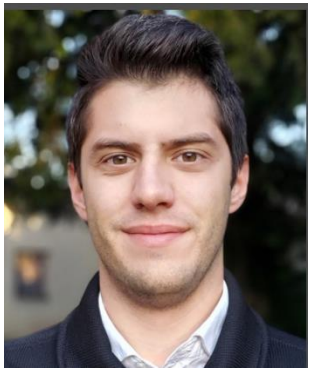
Lecturers

1. Prof. Manuel Roveri
   - HW-SW infrastructure, Cloud, MLaaS, Disks
   - Dipartimento di Elettronica, Informazione e Bioingegneria
   - manuel.roveri@polimi.it
   - http://roveri.faculty.polimi.it

2. Prof. Danilo Ardagna
   - Performance
   - Dipartimento di Elettronica, Informazione e Bioingegneria
   - danilo.ardagna@polimi.it
   - http://ardagna.faculty.polimi.it

Lecturers

3. Ing. Roberto Sala
   - Dependability
   - Dipartimento di Elettronica, Informazione e Bioingegneria
   - roberto.sala@polimi.it



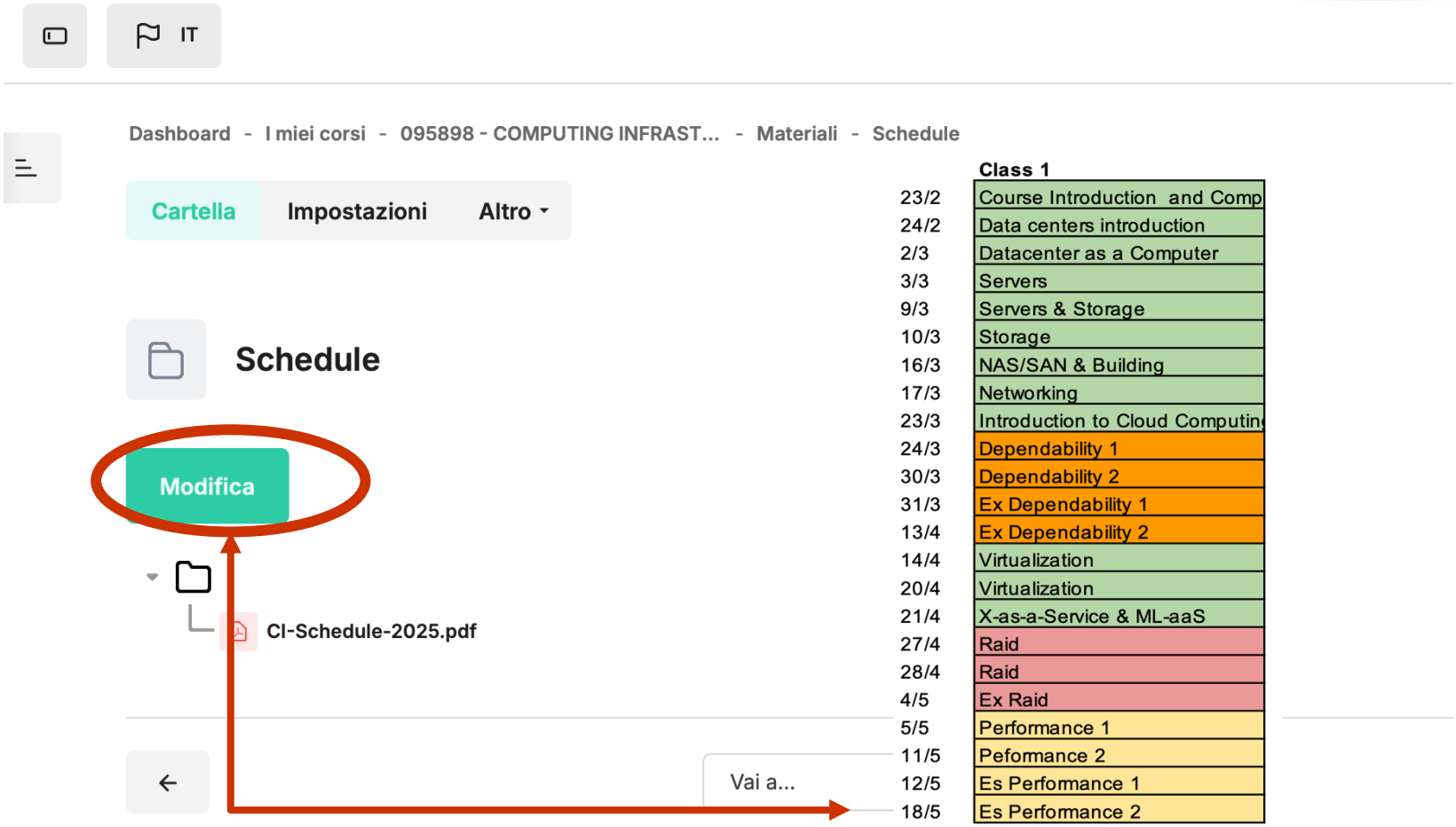
4. Prof. Marco Gribaudo
   - Dependability, disks, and performance exercises
   - Dipartimento di Elettronica, Informazione e Bioingegneria
   - marco.gribaudo@polimi.it

Course classrooms

	CS [E-P], GES[E-P], MUS[E-P],BIO, TEL[E-P], GEO
Monday	14:15-16:15 5.1.1
Tuesday	12:15-14:15 9.0.1

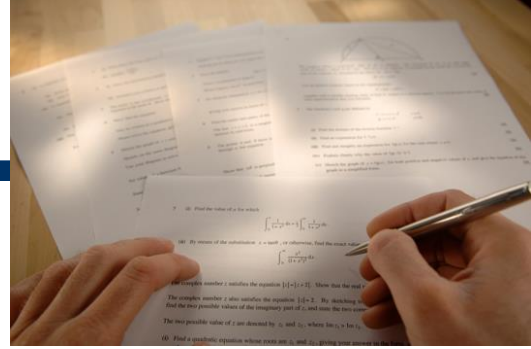
Schedule of the Course

- A detailed schedule is available on WeBeep



	Class 1
23/2	Course Introduction and Comp
24/2	Data centers introduction
2/3	Datacenter as a Computer
3/3	Servers
9/3	Servers & Storage
10/3	Storage
16/3	NAS/SAN & Building
17/3	Networking
23/3	Introduction to Cloud Computin
24/3	Dependability 1
30/3	Dependability 2
31/3	Ex Dependability 1
13/4	Ex Dependability 2
14/4	Virtualization
20/4	Virtualization
21/4	X-as-a-Service & ML-aaS
27/4	Raid
28/4	Raid
4/5	Ex Raid
5/5	Performance 1
11/5	Peformance 2
12/5	Es Performance 1
18/5	Es Performance 2
19/5	No class
25/5	Peformance 3
26/5	Es Performance 3

Evaluation



- The course will have a written exam
- The exam will consist in:
  - a set of exercises, i.e., simple problems to solve (similar to the ones that will be covered during the classes)
  - questions (True/False and open) dealing with more general topics covered by the course
- Closed book

- Exam will be, possibly, supported by Safe Browser
- **Mid-term exam** between

  April 15-18
  - if you pass mid term exam during the first call your exam will consist only of the second part of the course

Safe Exam Browser

SEB

https://safeexambrowser.org/news_en.html

Materials

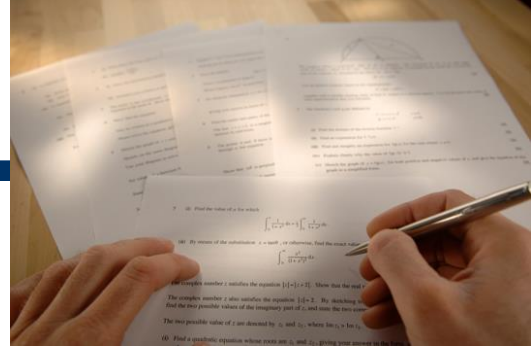
Slides of the course will be published on the WeBeep website

Materiali

Consulta i materiali forniti dal docente.

Cartella	**Schedule**
Cartella	**Slides**
Cartella	**Additional materials**
Cartella	**Past exams**

Evaluation



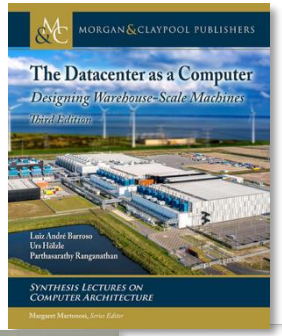
- Erasmus/mobility students: We follow the school directives (remote)

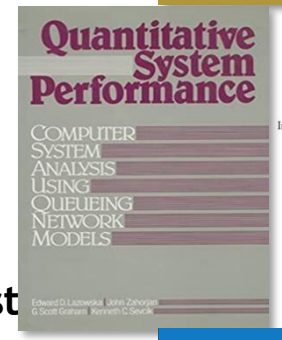
- Exercises and Exams will be made available
- Thesis about course topics will be made available
- Industry seminars

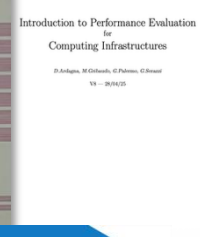
Bibliography

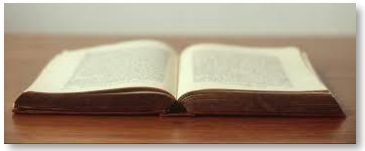
- Luiz André Barroso and Urs Hölzle, **The Datacenter as a Computer: Designing Warehouse-Scale Machines. 3rd Edition (2018)**
- Edward D. Lazowska, John Zahorjan,G. Scott Graham, Kenneth C. Sevcik, **Quantitative System Performance: Computer System Analysis Using Queueing Network Models.** (Ch.1-6) **+ Course Notes**
- Jerome H. Saltzer & M. Frans Kaashoek. **Fault Tolerance - Reliable Systems from Unreliable Components**

PDFs are freely available









Additional Material provided in WEBEEP

- **Further Readings:**
  - Caesar Wu and Rajkumar Buyya, **"Cloud Data Centers and Cost Modeling"**
  - James Smith and Ravi Nair, **"Virtual Machines - Versatile Platforms for Systems and Processes"**
  - Massimo Lazzaroni Loredana Cristaldi Lorenzo Peretto Paola Rinaldi Marcantonio Catelani **"Reliability Engineering: Basic Concepts and Applications in ICT"**
  - Giuseppe Serazzi, **"Performance Engineering: Learning Through Applications Using JMT"**